IV. L^1 space, ~~Differentiation and Integration~~

1. L^1 space

Def IV.1.1. f is a complex-valued function on $\mathbb{R}^d$. say $f(x) = u(x) + iv(x)$

f is Lebesgue integrable iff $|f(x)| = (u^2+v^2)^{\frac{1}{2}}$ is integrable

Then define $\int f(x)\,dx = \int u(x)\,dx + i\int v(x)\,dx$

Def IV.1.2. Given any integrable function f on $\mathbb{R}^d$ We define $\|f\|_{L^1} = \int_{\mathbb{R}^d} |f(x)|\,dx$

Def IV.1.3 $f \sim g$ iff $f = g$ a.e. x. All equivalent classes of integrable functions on $\mathbb{R}^d$ is called the space $L^1(\mathbb{R}^d)$

Prop IV.1.4. Let $f, g \in L^1(\mathbb{R}^d)$, then $d(f,g) = \|f-g\|_{L^1(\mathbb{R}^d)}$ defines a metric on $L^1(\mathbb{R}^d)$

Moreover, $L^1(\mathbb{R}^d)$ is complete in its metric (Riesz-Fischer)

Proof: $\forall \{f_n\}_{n\geq 1}$ Cauchy sequence in the norm. that is $\|f_n - f_m\| \to 0$, $n,m \to \infty$

$\exists \{f_{n_k}\}_{k\geq 1}$, s.t. $\|f_{n_{k+1}} - f_{n_k}\| \leq 2^{-k}$ $\forall k \geq 1$.

Let $f(x) = f_{n_1}(x) + \sum_{k=1}^{\infty} (f_{n_{k+1}}(x) - f_{n_k}(x))$. $g(x) = |f_{n_1}(x)| + \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)|$

Then $\int |f_{n_1}| + \sum_{k=1}^{\infty} \int |f_{n_{k+1}} - f_{n_k}| \leq \int |f_{n_1}| + \sum_{k=1}^{\infty} 2^{-k} < \infty$.

By M.C.T, we have g is integrable $\Rightarrow |f| \leq g \Rightarrow f$ is integrable.

$\Rightarrow f_{n_k}(x) \to f(x)$ a.e. x Since $|f - f_{n_k}| \leq g$. $\forall k \geq 1$.

By D.C.T, we have $\lim_{k\to\infty} \int |f - f_{n_k}| = \int \lim_{k\to\infty} |f - f_{n_k}| = 0 \Rightarrow \|f - f_{n_k}\|_{L^1} \to 0$ $(k \to \infty)$

Now $\forall \varepsilon > 0$ $\exists N$ s.t. $\forall n, m > N$. $\|f_n - f_m\| < \frac{\varepsilon}{2}$. choose $n_k > N$ s.t. $\|f - f_{n_k}\| < \frac{\varepsilon}{2}$

$\Rightarrow \forall n > N$. $\|f_n - f\| \leq \|f_n - f_{n_k}\| + \|f_{n_k} - f\| < \varepsilon$. thus $f_n \xrightarrow{L^1} f$ and $f \in L^1(\mathbb{R}^d)$ $\square$.

Cora IV.1.5. In the above proof, we can have that: If $\{f_n\}_{n\geq 1}$ converges to f in $L^1(\mathbb{R}^d)$, then $\exists$ sequence $\{f_{n_k}\}_{k\geq 1}$ s.t. $f_{n_k}(x) \to f(x)$ a.e. x.

Rmk. $\|f_n - f\|_{L^1} \to 0$ ~~don't~~ doesn't mean that $f_n(x) \to f(x)$ a.e. x.

Thm IV.1.6. The following families of functions are dense in $L^1(\mathbb{R}^d)$:

(i) The simple functions

(ii) The step functions

(iii) The continuous functions of compact support.

Proof: (i) Using pointwise convergence: $\exists \{\varphi_k\}$. $\varphi_k \nearrow f$, Then by M.C.T, we have (sketched). $\int \varphi_k \to \int f \Rightarrow \|\varphi_k - f\|_{L^1} \to 0$

(ii) W.L.O.G. let $f = \chi_E$., $m(E) < \infty$. Using the previous approximation theory.

(iii) W.L.O.G. let $f = \chi_E$ E is a rectangle. By taking the product of f_k $(1 \leq k \leq d)$ on $\mathbb{R}$. $\square$.

2. Translation and continuity Let $f_h(x) = f(x-h)$

Prop IV.2.1 Suppose $f \in L^1(\mathbb{R}^d)$ Then $\|f_h - f\|_{L^1} \to 0$ as $h \to 0$

proof. (sketched) Using the continuous function with compact support to approximate f $\square$

3. Fubini's theorem.

Def IV.3.1. (1) Let $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. If f is a function in $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$.

the slice of f corresponding to $y \in \mathbb{R}^{d_2}$ is $f^y(x) = f(x, y)$

(2) $E \subseteq \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ $E^y = \{x \in \mathbb{R}^{d_1} \mid (x,y) \in E\}$ $E_x = \{y \in \mathbb{R}^{d_2} \mid (x,y) \in E\}$

(Fubini).

Thm IV.3.2. Suppose $f(x,y)$ is integrable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. Then for <u>almost</u> every $y \in \mathbb{R}^{d_2}$:

(i) f^y is integrable on $\mathbb{R}^{d_1}$.

(ii) $\int_{\mathbb{R}^{d_1}} f^y(x)\,dx$ is integrable on $\mathbb{R}^{d_2}$.

(iii) $\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x,y)\,dx \right) dy = \int_{\mathbb{R}^{d_1}} \left(\int_{\mathbb{R}^{d_2}} f(x,y)\,dy \right) dx = \int_{\mathbb{R}^d} f(x,y)$.

Proof(sketched): Let $\mathcal{F}$ be the set of integrable functions on $\mathbb{R}^d$ satisfying (i), (ii), (iii).

Step 1. Any finite linear combination of functions in $\mathcal{F}$ also belongs to $\mathcal{F}$

Step 2. $\{f_k\}_{k\geq 1} \subseteq \mathcal{F}$ $f_k \nearrow f$ or $f_k \searrow f$. where f_k is integrable, then $f \in \mathcal{F}$

Step 3. $\chi_E \in \mathcal{F}$ $m(E) < \infty$, E is a G_δ set.

Step 4. $\chi_E \in \mathcal{F}$ $m(E) = 0$

Step 5. $\chi_E \in \mathcal{F}$ $m(E) < \infty$

Step 6. $\forall f \in L^1(\mathbb{R}^d)$ $f = f^+ - f^-$ W.L.O.G (1) $f \geq 0 \Rightarrow \exists$ simple $\{\varphi_k\}_{k\geq 1}$. $\varphi_k \nearrow f$.

By (5) & (1) we have $\varphi_k \in \mathcal{F}$. By (2) we have $f \in \mathcal{F}$ $\square$.

Rmk. If f is simply measurable, $\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x,y)\,dx \right) dy = \int_{\mathbb{R}^{d_1}} \left(\int_{\mathbb{R}^{d_2}} f(x,y)\,dy \right) dx$ in the extend sense.

proof: Using the bounded situation to approximate the general case $\square$.